

Product Application Guide

ForgeCore Industries — Applications Engineering

Selecting the Right Crucible for Your Application

Choosing the correct crucible material is critical to furnace life, melt quality, and operational cost. This guide walks through the decision factors and presents our recommendations for common industrial applications.

Decision Factors

1. Metal or alloy being melted (chemistry compatibility)
2. Maximum melting temperature required
3. Furnace atmosphere (oxidising, neutral, reducing, inert)
4. Charge weight and pour frequency
5. Cost-per-heat vs lifespan trade-off

Application Recommendations

Iron and Steel Melting

Recommended product: FC-45

Silicon carbide offers excellent thermal shock resistance for intermittent furnace cycles typical in iron foundries. Service life 300-500 heats under normal conditions. Recommended for batch sizes 50-500 kg.

Copper and Copper Alloy Melting

Recommended product: FC-50

High-purity graphite is the industry standard for copper melting due to non-wetting behaviour and minimal carbon pickup. Use under protective atmosphere when melt exceeds 1,200°C.

Aluminium Die-Casting Foundries

Recommended product: FC-72

Clay-graphite composite is the most economical choice for high-volume aluminium operations. Avoid in furnaces exceeding 1,300°C — consider FC-45 for higher temperatures.

Precious Metals (Gold, Silver, Platinum)

Recommended product: FC-90

Zircon's chemical inertness minimises metal loss and contamination. Essential for high-purity alloys and jewellery-grade melts. Higher cost is justified by metal value preservation.

Zinc and Zinc Alloys

Recommended product: FC-72

Lower operating temperatures of zinc allow the economical clay-graphite composite. Service life commonly exceeds 1,000 heats.

Magnesium Allting

Recommended product: FC-90

Zircon crucibles with protective flux coating are mandatory for magnesium due to high reactivity. Special handling required — see Safety Guidelines document.

Laboratory and R&D Use

Recommended product: FC-AX

Alumina ceramic crucibles provide the highest purity environment for analytical and research applications. Limited to non-ferrous, non-reactive metals.

Common Application Mistakes

Avoid these frequent errors that significantly reduce crucible life:

- Exceeding rated maximum temperature (even by 50°C halves lifespan)
- Charging cold metal into a hot crucible without pre-warming the charge
- Using inappropriate tongs or handling equipment that damages crucible walls
- Mixing crucible materials in the same furnace (carbon pickup in oxide crucibles)
- Inadequate pre-heating on initial use (follow the staged heating procedure)

Pre-Heating Procedure (All Crucibles)

Before first use and after any extended storage period, all crucibles must be conditioned by a controlled pre-heat to remove residual moisture and stress-relieve the structure:

- Load crucible into cold furnace
- Heat at 50°C/hour to 200°C, hold 2 hours
- Heat at 100°C/hour to 600°C, hold 1 hour
- Continue to operating temperature at 150°C/hour

Total conditioning time: 8-12 hours. Failure to condition results in thermal shock cracking on first heat-up.